

**SECTION B**

*Answer all questions in the spaces provided.*

7. (a) Ammonia is an example of a weak base. Describe what is meant by the term *weak base*. [1]

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- (b) A mixture of ammonia and ammonium chloride in aqueous solution can be used as a basic buffer solution. Explain what is meant by a *buffer solution* and how this mixture can act as a buffer solution. [3]

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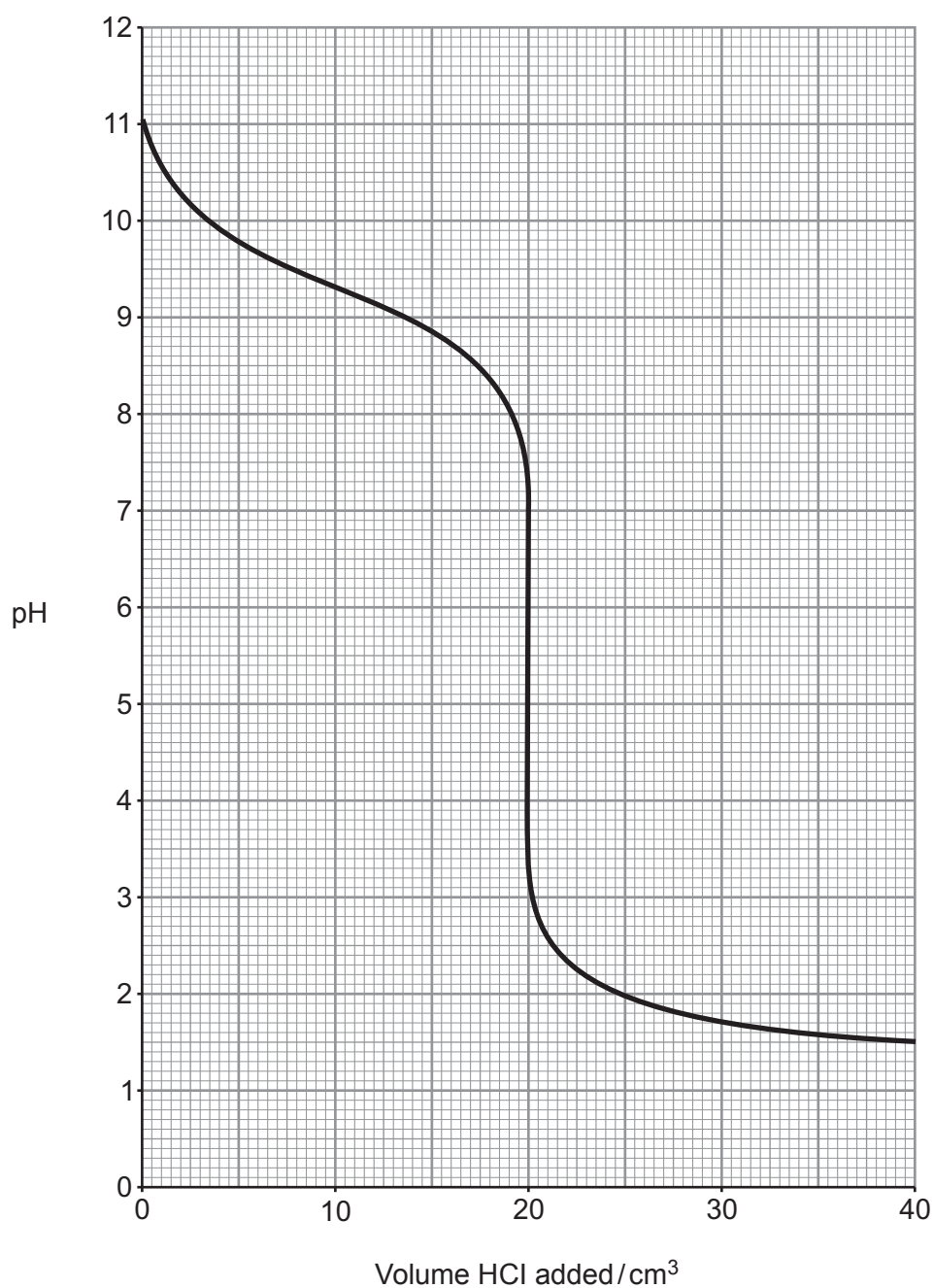
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- (c) The pH curve for titration of a solution of ammonia against hydrochloric acid of an equal concentration is given below.



- (i) Calculate  $K_a$  for the ammonium ion.

[2]

$K_a = \dots\dots\dots \text{mol dm}^{-3}$



- (ii) Select an appropriate indicator for this titration from the list below, giving a reason for your answer. [1]

| Indicator        | pH range   |
|------------------|------------|
| bromothymol blue | 6.0 – 7.6  |
| 4-nitrophenol    | 5.0 – 7.0  |
| methyl yellow    | 2.9 – 4.0  |
| phenolphthalein  | 8.2 – 10.0 |

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